



Contents lists available at ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

Mixed butanols addition to gasoline surrogates: Shock tube ignition delay time measurements and chemical kinetic modeling

Abdullah S. AlRamadan^{a,*}, Jihad Badra^{a,*}, Tamour Javed^b, Mohammed Al-Abbad^b,
Nehal Bokhumseen^a, Patrick Gaillard^a, Hassan Babiker^a, Aamir Farooq^b, S. Mani Sarathy^b

^a Saudi Aramco Research and Development Center, Fuel Technology R&D Division, Dhahran 31311, Saudi Arabia

^b Clean Combustion Research Center, Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal 23955, Saudi Arabia

ARTICLE INFO

Article history:

Received 15 April 2015

Revised 26 July 2015

Accepted 26 July 2015

Available online xxx

Keywords:

Butanols

Octane boosters

Shock tube

Ignition delay times

Chemical kinetics

Toluene primary reference fuels

ABSTRACT

The demand for fuels with high anti-knock quality has historically been rising, and will continue to increase with the development of downsized and turbocharged spark-ignition engines. Butanol isomers, such as 2-butanol and tert-butanol, have high octane ratings (RON of 105 and 107, respectively), and thus mixed butanols (68.8% by volume of 2-butanol and 31.2% by volume of tert-butanol) can be added to the conventional petroleum-derived gasoline fuels to improve octane performance. In the present work, the effect of mixed butanols addition to gasoline surrogates has been investigated in a high-pressure shock tube facility. The ignition delay times of mixed butanols stoichiometric mixtures were measured at 20 and 40 bar over a temperature range of 800–1200 K. Next, 10 vol% and 20 vol% of mixed butanols (MB) were blended with two different toluene/n-heptane/iso-octane (TPRF) fuel blends having octane ratings of RON 90/MON 81.7 and RON 84.6/MON 79.3. These MB/TPRF mixtures were investigated in the shock tube conditions similar to those mentioned above. A chemical kinetic model was developed to simulate the low- and high-temperature oxidation of mixed butanols and MB/TPRF blends. The proposed model is in good agreement with the experimental data with some deviations at low temperatures. The effect of mixed butanols addition to TPRFs is marginal when examining the ignition delay times at high temperatures. However, when extended to lower temperatures ($T < 850$ K), the model shows that the mixed butanols addition to TPRFs causes the ignition delay times to increase and hence behaves like an octane booster at engine-like conditions.

© 2015 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

1. Introduction

Over the last few decades, spark-ignition (SI) engines have encountered great improvements in fuel economy and emissions, driven by stringent environmental regulations [1]. The automotive industry has worked extensively to develop engines that utilize existing fuels (e.g., gasoline), while increasing the engine performance and durability. This development is expected to continue according to future projections, which anticipate a movement towards reducing fuel consumption of SI engines through downsizing and turbocharging (i.e., boosting) [2–4]. Downsizing and turbocharging can improve engine thermal efficiency by reducing throttling losses, as the engine will operate mostly at full throttle. Also, the downsized engine has lower weight and volume causing the frictional (mechanical) losses to decrease. In addition, the engine is expected to operate at higher loads due to turbocharging, which also contributes to improve the

overall efficiency of the engine. However, pursuing such technologies requires fuels with more resistance to auto-ignition because the high-load operating conditions of the engine will potentially increase the possibility engine knock [5,6]. Therefore, knock occurrence is the major constraint that prevents SI engines from reaching their full potential.

Kalghatgi [5,6] defines knock as the metallic ringing noise of engine, caused by the auto-ignition of end gas located behind the propagating flame, which, in turn, creates pressure waves in the combustion chamber. This abnormal phenomenon is undesirable because knock can potentially cause severe damage to the piston's head and reduce the efficiency of the engine, as the mixture auto-ignites before reaching the optimum conditions. Hence, fuels used in SI engines are classified by octane ratings, which indicate the propensity of fuels to auto-ignite. The octane rating of a certain fuel is given by the Research Octane Number (RON) and the Motor Octane Number (MON). The RON and MON ratings of a fuel can be obtained from standard test methods in Cooperative Fuel Research (CFR) engine [7,8].

As the demand for fuels with high anti-knock quality is rising, there is a significant interest in blending agents (i.e., octane boosters)

* Corresponding author.

E-mail address: jihad.badra@aramco.com (J. Badra).

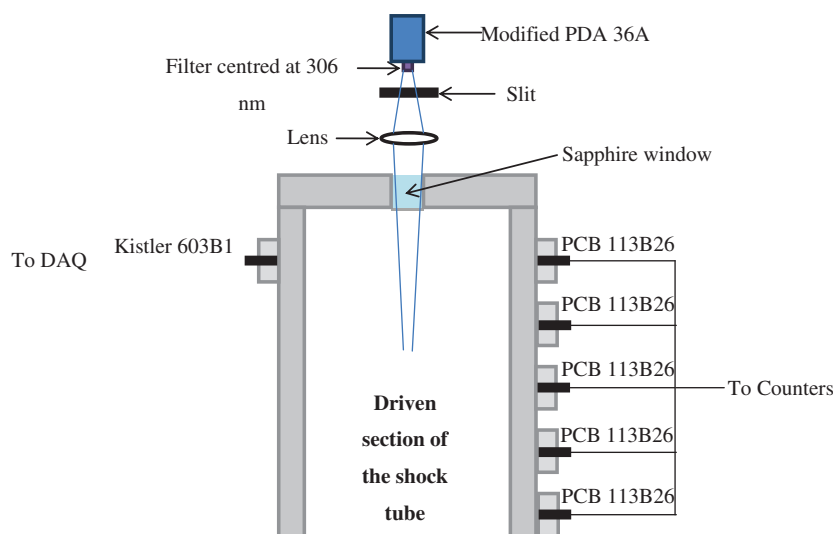


Fig. 1. Experimental setup for ignition delay time measurements.

with conventional gasoline fuels. Alcohols with 1–4 carbon atoms have been proposed for blending with gasoline due to their high octane number [9]. Amongst the alcohols, the butanol isomers offer advantages over ethanol, the most widely used biofuel globally. Firstly, the butanol isomers have a higher energy density than ethanol, resulting in lower volumetric fuel consumption [10]. In addition, unlike ethanol, butanol has a lower tendency to mix with water, which enables higher volume fractions to be blended with gasoline and transport it using existing pipelines [11,12].

There are four butanol isomers: n-butanol, 2-butanol, iso-butanol and tert-butanol. All of the isomers can be produced through biological pathways apart from tert-butanol, which is a petroleum derived product [9,13]. Combustion of the butanol isomers has been studied extensively in shock tubes [14–24]. This greatly contributed to the development of many chemical kinetic models [9,18,22,24–29]. These chemical kinetic models provide a detailed description of the fuel ignition characteristics and hence a better understanding of the fuel/engine interactions through engine-combustion simulations.

The present work focuses on studying the combustion of mixed butanols (MB), a mixture of 68.8% 2-butanol and 31.2% tert-butanol by volume, blended with ternary component gasoline surrogates. The production of combustible mixed butanols from mixed butenes has been patented by the Saudi Aramco [30,31]. The potential use of mixed butanols as oxygenated fuel additives has been studied in DI single cylinder engine by Niass et al. [32], wherein the authors found that MB addition to a commercial RON 97 gasoline resulted in improved knock resistance, higher indicated mean effective pressure (IMEP), and lower particulate emissions. However, no prior effort has been made to investigate mixed butanols combustion at a fundamental level. In this study, a chemical kinetic model has been developed to understand the effects on combustion when adding mixed butanols to gasoline surrogate mixtures. The mechanism is compared against high pressure shock tube ignition delay time measurements for pure mixed butanols and blends of mixed butanols with iso-octane/n-heptane/toluene mixtures.

2. Experimental procedure

In this study, ignition delay times were measured using the high-pressure shock tube (HPST) facility at King Abdullah University of Science and Technology (KAUST). The experimental setup has been detailed elsewhere [33] and only brief description is given here. The HPST is made from high purity stainless steel with inner diameter

of 10 cm, driven section of 6.6 m length and driver section can be varied in length (maximum of 6.6 m) depending on the desired test time. The driver and driven sections are separated by a mid-section designed to hold two pre-scored diaphragms. Aluminum diaphragms with various thicknesses and scorings were used depending on the desired pressure behind reflected shock wave (P_5). The double diaphragms technique was implemented to achieve a better control of diaphragm rupturing at the desired conditions. Moreover, the facility has two stainless steel mixing vessels attached to the driven section for gaseous air/fuel mixtures preparation. The mixing vessels and the shock tube are vacuumed using turbomolecular pumps to remove impurities. The driven section of the HPST, the mixing vessels and the connecting tubes can be heated to 100 °C to avoid fuel condensation. Five equally separated PCB 113B26 piezoelectric pressure transducers (PZT) are placed axially in the last 1.5 m of the driven section to measure incident shock speed. The pressure time-history of the activity inside the shock tube is recorded using Kistler 603B1 PZT located 1 cm away from the HPST end-wall. OH* chemiluminescence, light emissions at 306 nm, is monitored using a modified PDA36A detector via Sapphire window in the end-wall. Schematic of the experimental setup in the driven section is shown in Fig. 1.

Gaseous mixtures were prepared by injecting liquid fuel into two evacuated and heated (100 °C) stainless steel mixing vessels. Fuel partial pressure was kept at two-thirds or less of the maximum saturation vapor pressure. Oxygen (purity 99.999%) and nitrogen (purity 99.999%), supplied by Abdullah Hashim Gases, were mixed with the fuel in the mixing vessels to reach desired equivalence ratio. A magnetically driven stirrer was used for at least one hour to mix the fuel vapor with air and produce homogenous mixture. When filling the driven section with air/fuel mixture, butanol has the tendency to adsorb to the shock tube walls. To minimize the adsorption effect, a passivation technique was performed in which the driven section was overfilled by the air/fuel mixture beyond the initial pressure. After that, the excess gas was evacuated from the shock tube until optimal pressure was achieved. The passivation technique is adopted from Stranic et al. [14] shock tube study on butanol isomers. The driver section was pressurized with either pure helium or helium–nitrogen mixture based on the required test time. Refer to [34,35] for further details on the driver gas tailoring technique to extend shock tube test times. The mid-section was filled with the driver gas at half the pressure of the driver section. Then, the shock wave was initiated by venting the middle section, which causes the diaphragms to rupture due to the large pressure gradient between the driver and driven sections.

Table 1
Octane ratings of fuels used in this study.

	RON	MON	Ref.
2-butanol	105	93	[47]
Tert-butanol	107	94	[48]
Iso-octane	100	100	–
n-Heptane	0	0	–
Toluene	120	109	[49]

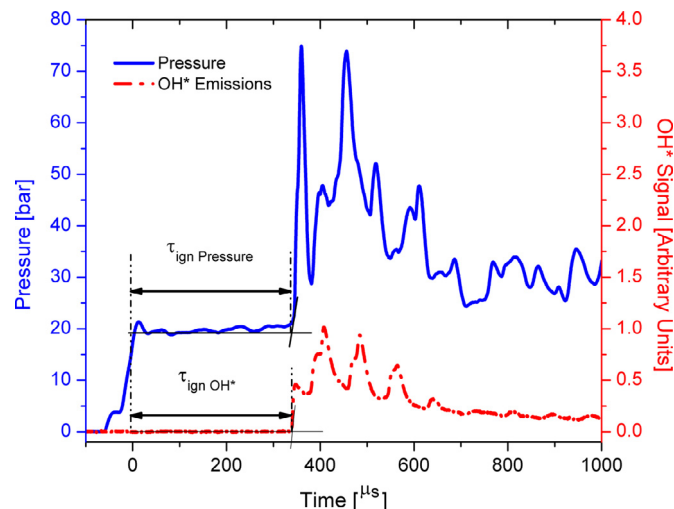


Fig. 2. Determination of ignition delay time from OH* chemiluminescence and pressure signals, $P_5 = 19.8$ bar $T_5 = 1104$ K, 68.8% 2-butanol/31.2% tert-butanol/air. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article).

The temperature and pressure (T_5 and P_5) behind the reflected shock wave at which the air/fuel mixture reacts were calculated using the linearly extrapolated shock speed and one-dimensional shock-jump equations. All ignition delay times (τ_{ign}) in this study were determined by calculating the time interval between the arrival of the reflected shock wave and the steepest increase in the OH* emissions (Fig. 2). Another method to calculate the ignition delay time is to extrapolate the steepest increase in the pressure signal (blue trace in Fig. 2). Ignition delay times obtained with these two methods agree within $\pm 2\%$.

Mixed butanols are prepared using 2-butanol ($\geq 99\%$) and tert-butanol ($\geq 99\%$) obtained from Sigma Aldrich. Toluene ($\geq 99.8\%$), isooctane ($\geq 99\%$) and n-heptane ($\geq 99\%$) are also purchased from

Sigma Aldrich. The octane ratings of the pure fuels used in this study are summarized in Table 1. The CFR engine facility available at Saudi Aramco was utilized to measure to the octane numbers of mixed butanols. Mixed butanols have RON and MON of 106.7 and 94, respectively. Further details on the concentrations of n-heptane, iso-octane, and toluene are provided later in the manuscript.

3. Chemical kinetics model development

A detailed chemical kinetics model that includes 2-butanol, tert-butanol, n-heptane, iso-octane and toluene mechanisms is required to perform ignition delay time simulations of the mixed butanols blended with ternary gasoline surrogates. Thus, the recently published C_1 – C_5 alcohols mechanism by Sarathy et al. [9] was selected as the primary mechanism, which utilizes the AramcoMech 1.3 C_0 – C_2 base chemistry [36] and the butanol isomers combustion model from [20,25]. The alcohols combustion chemistry model was merged with gasoline surrogates model developed by Mehl et al. [37], which contains n-heptane, iso-octane and toluene sub-chemistries. The species names were unified between the two mechanisms and any duplicate thermodynamic data and reactions were adopted from the alcohols model. Table 2 summarizes the content of each kinetic model, and the range of experimental conditions at which they are tested.

Several minor modifications to the rate constants of 2-butanol and tert-butanol specific reactions were made to improve the simulated ignition delay times. The H-abstraction from 2-butanol and tert-butanol by OH reaction rate coefficients was taken from the measurements of McGillen et al. [38] and has been implemented in the combined mechanism. The rate of H-atom abstraction by HO_2 from the alpha-site of 2-butanol was increased by a factor of 2, which is within the uncertainty of the original estimation [25]. The modified reactions and their Arrhenius parameters are presented in Table 3.

The present model is compared against the ignition delay time data using the adiabatic homogenous reactor model in CHEMKIN PRO [39] with an imposed volume history determined using the variable volume method (VTIM) proposed by Chaos and Dryer [40] to account for the experimentally measured pressure gradient of $dP/dt(1/P_0) = 5\%/ms$.

4. Results, discussions and analysis

4.1. Model comparisons for pure components

Initially, the proposed chemical kinetic model was compared against ignition delay time measurements reported in literature for

Table 2
Summary of the mechanisms merged in this study.

	Sarathy et al. [9] alcohols mechanism	Mehl et al. [37] gasoline surrogates mechanism
Species included	C_1 – C_5 alcohols	n-Heptane, iso-octane, 1-hexene, and toluene
Tested range	$\phi = 0.5$ – 2.0 750–1600 K 1.2–80 bar	$\phi = 1$ 650–1200 K 3–50 atm

Table 3
Summary of the mechanisms merged in this study.

Reaction	C–H type	A (cm,mol,s)	B	E_a (cal/mol)
$sC_4H_9OH + OH \rightleftharpoons sC_4H_8OH + H_2O$	α	$2.960e+05$	2.20	$-2.20e+03$
$sC_4H_9OH + OH \rightleftharpoons sC_4H_8OHm + H_2O$	β (methyl)	$8.280e+04$	2.41	$-1.18e+03$
$sC_4H_9OH + OH \rightleftharpoons sC_4H_8OH-2 + H_2O$	β (ethyl)	$5.850e+04$	2.52	$-1.30e+03$
$sC_4H_9OH + OH \rightleftharpoons sC_4H_8OH-3 + H_2O$	gamma	$8.280e+04$	2.41	$-1.18e+03$
$sC_4H_9OH + OH \rightleftharpoons sC_4H_9O + H_2O$	OH	$1.260e+06$	2.00	$4.57e+01$
$tC_4H_9OH + OH \rightleftharpoons tC_4H_8OH + H_2O$	β	$8.070e+06$	2.00	$4.51e+02$
$tC_4H_9OH + OH \rightleftharpoons tC_4H_9O + H_2O$	OH	$1.260e+06$	2.00	$-4.57e+01$
$sC_4H_9OH + HO_2 \rightleftharpoons sC_4H_8OH-1 + H_2O_2$	α	$2.500e-05$	5.26	$7.47e+03$

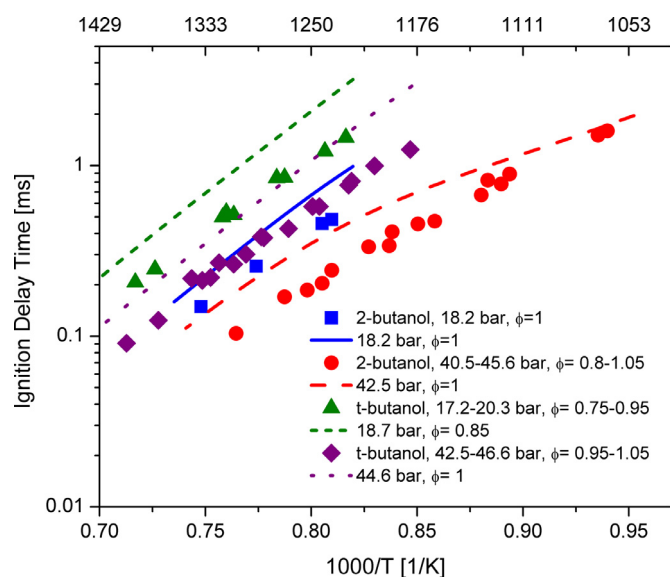


Fig. 3. Measured ignition delay times in shock tube from Stranic et al. [14] (symbols) and simulations (lines) for 2-butanol and tert-butanol.

pure 2-butanol and tert-butanol. This comparison is a check to ensure that kinetic modeling of these pure components is not affected by merging sub-mechanisms and their subsequent cross-reactions. Fig. 3 compares the results from the simulations using the merged kinetic model with high-pressure shock tube experiments performed by Stranic et al. [14]. The proposed kinetic model captures the overall slope of the ignition delay curves for both 2-butanol and tert-butanol, and the model captures pressure-dependency of ignition delay times. At the temperatures and pressures presented in Fig. 3, both simulations and experiments show that the ignition delay times for 2-butanol are shorter than tert-butanol. Hence, it can be concluded that 2-butanol has overall higher reactivity than tert-butanol at the studied conditions. The model, however, over-predicts the ignition delay times for both fuels at the presented operating conditions. Analysis of the chemical kinetic model revealed that the decreased reactivity is due high temperature C_0 – C_3 reactions in the base chemistry. The global reactivity is sensitive to the reaction $H+O_2 \rightleftharpoons O+OH$ and oxidation of ethyl and propyl radical, while not sensitive to the alcohol sub-chemistry.

It is also useful to compare the proposed model to low temperature/high pressure (engine-like) experimental conditions. Weber et al. [41,42] performed such experiments for butanol isomers using rapid compression machine (RCM), which enables studying low temperature reactivity of fuels at well-controlled conditions. A comparison between experimental and simulated ignition delay times for stoichiometric 2-butanol/air and tert-butanol/air mixtures at $P = 15$ atm is shown in Fig. 4. RCM experimental results, in general, are affected by heat losses through the reactor walls. Consequently, the reactivity of the constant volume adiabatic simulations is significantly higher than the RCM experimental data as can be seen from Fig. 4. Weber et al. [41] performed nonreactive experiments in which the oxidizer was replaced by nitrogen to monitor the heat loss. The non-reactive volume profiles for each data point were created from the specific heat value of the mixture and adiabatic relations. These volume profiles were fed into CHEMKIN-PRO to account for heat losses and the results improved slightly for both 2-butanol and tert-butanol, as demonstrated in Fig. 4.

4.2. Ignition delay time simulations of mixed butanols

Ignition delay times of binary mixture of 68.8% 2-butanol and 31.2% tert-butanol, which are the constituents of mixed butanols,

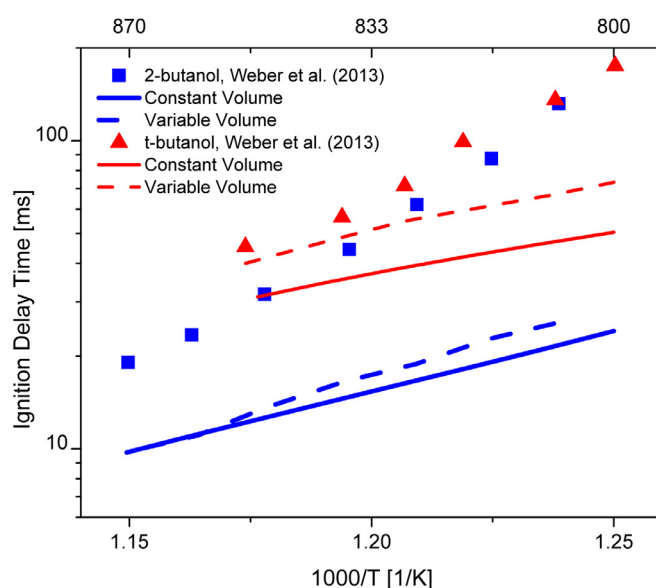


Fig. 4. Ignition delay time data in rapid compression machine obtained from Weber et al. [43] (symbols) and simulations (lines) for 2-butanol and tert-butanol/air, $P = 15$ atm and $\phi = 1$.

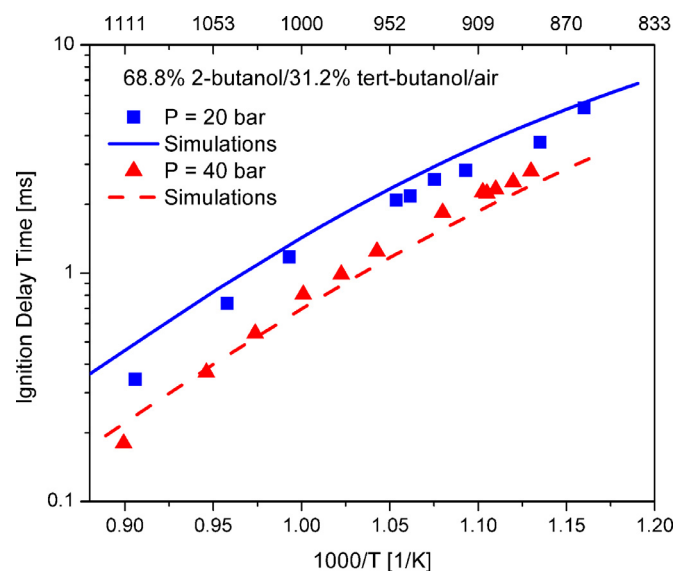


Fig. 5. Shock tube ignition delay time measurements of mixed butanols (symbols) and simulations (lines) at $\phi = 1$.

were measured for stoichiometric fuel/air mixtures at 20 and 40 bar over a temperature range of 800–1200 K. As mentioned earlier, the pressure behind the reflected shock wave increases gradually due to boundary layer effects. This increase in pressure results in an increase of temperature. This effect must be taken into consideration for relatively long test time experiments. The pressure increase is converted to volume profile using:

$$V_{\text{relative}}(t) = \left(\frac{P(t)}{P_{\text{initial}}} \right)^{\frac{1}{\gamma}} \quad (1)$$

where γ is the specific heat ratio. With this relationship, volume profiles were generated and used in CHEMKIN-PRO to perform ignition delay time simulations. Figure 5 shows the comparison between the simulated and measured ignition delay times for mixed butanols/air stoichiometric mixture. The proposed model is in good agreement with the experimental results and captures the overall slope of the ignition delay curves with some deviation at low temperature data of

Table 4
Specifications of the base gasoline analyzed in this study.

	Gasoline-A	Gasoline-B
Paraffins (vol%)	19.0	24.6
Iso-paraffins (vol%)	34.0	35.0
Aromatics (vol%)	42.8	33.7
Olefins (vol%)	1.8	0
Naphthenes (vol%)	2.5	4.8
Oxygenates (vol%)	0	0
RON	90.0	84.6
MON	81.7	79.3

Table 5
Specifications of the TPRF mixtures tested in this study.

	TPRF-A	TPRF-B
n-Heptane (vol%)	21.9	22.9
Iso-octane (vol%)	28.7	48.3
Toluene (vol%)	49.4	28.8
Calculated RON [43]	90.0	84.6
Calculated MON [43]	81.7	79.3

40 bar experiments. In addition, both simulations and experiments predict an increase in reactivity when pressure increases from 20 to 40 bar.

4.3. Addition of mixed butanols to toluene/n-heptane/iso-octane (TPRF) mixtures

After comparing the chemical kinetic model for mixed butanols with experiments, the ignition behavior of mixed butanols blended with gasoline surrogates was examined. The surrogates used in this work are labeled as toluene primary reference fuels (TPRF). The TPRF surrogates consist of linear paraffins (n-heptane), branched paraffins (isooctane) and aromatics (toluene). The two TPRF mixtures are generated from a well-validated non-linear by mole blending rule, developed by Kalghatgi et al. [43] that predicts the octane rating of TPRF surrogates. The two base gasoline fuels that TPRF surrogates represent are the RON 90 and RON 84.6 Saudi Arabian gasoline fuels without the addition of octane boosters. Table 4 below presents the detailed composition of the base gasoline fuels. The composition of TPRF surrogate is given in Table 5.

Ignition delay time experiments were performed for TPRF A and B mixtures with 10 vol% and 20 vol% of mixed butanols at 20 bar and temperature range of 890–1150 K. Figure 6 presents the shock tube experimental data. The chemical kinetic model does a good job of capturing the experimental data. Over the studied temperature range, ignition delay times of the two fuel blends are quite similar, detecting a slight increase in reactivity with mixed butanol addition.

To further explore the effect of adding mixed butanols to TPRF mixtures, ignition delay times are calculated and plotted in Fig. 7 for TPRF mixture, mixed butanols and the two blends with 10% and 20% mixed butanols. At temperatures higher than 850 K, mixed butanols are slightly more reactive than TPRF-A. The opposite trend is observed at temperatures below 850 K and, hence, mixed butanols behave like an octane booster by delaying ignition at lower temperatures. When blending mixed butanols with TPRF-A, the ignition delay times decrease for temperatures above 820 K and increase for temperature below 820 K. However, the differences in the ignition delay times are marginal. In case of blending mixed butanols with TPRF-B, the same trends are observed with the cross-over temperature being 850 K. Also, the differences in the ignition delay times are more pronounced when blending mixed butanols with TPRF-B compared to TPRF-A. A possible explanation for this behavior is that the octane rating of TPRF-B (RON = 84.6) is lower than TPRF-A (RON = 90.0) and, hence, the effect of blending the high-octane agent, such as mixed

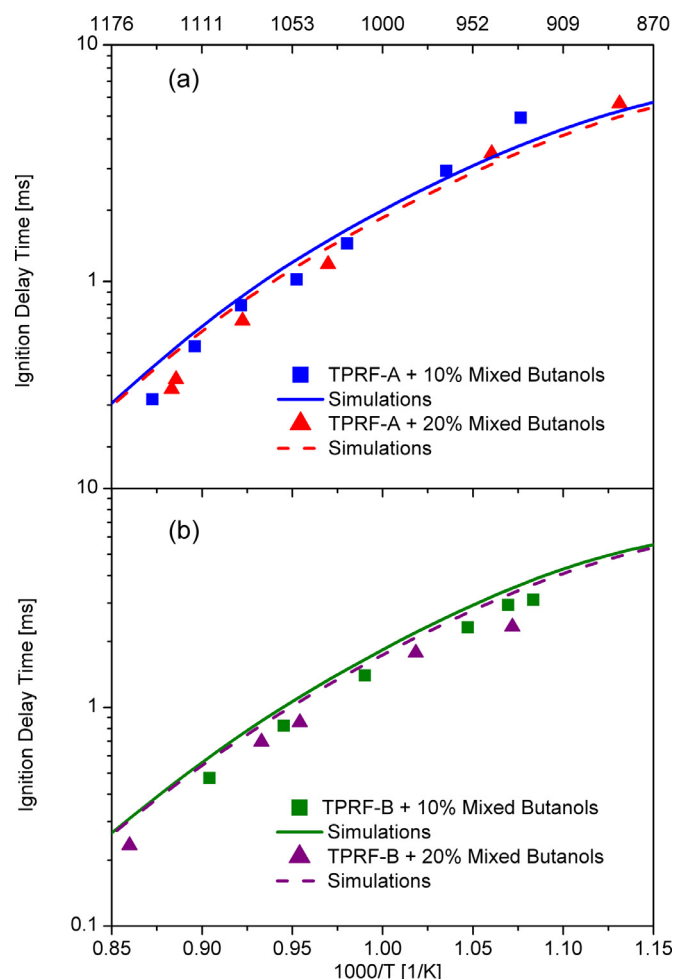


Fig. 6. Shock ignition delay time measurements of mixed butanols + toluene/isooctane/n-heptane mixtures at $P = 20$ bar, $\phi = 1$, (a) TPRF-A and (b) TPRF-B.

butanols, is more significant. Another possible explanation to the trend is that TPRF-B has higher iso-paraffinic content than TPRF-A which makes the ignition delay times of pure TPRF-B experience stronger negative temperature coefficient (NTC) reactivity compared to pure TPRF-A. The temperatures (i.e., 820 and 850 K) at which mixed butanols start acting as an octane booster agree well with the values reported in Mehl et al. [44] and Sarathy et al. [45]. Mehl et al. [44] developed a correlation between anti knock index (AKI) and ignition delay times at 825 K and 25 atm, while Sarathy et al. [45] reported that the ignition delay times at 835 K and 20 atm correlate with research octane numbers. All the measured ignition delay times along with their respective operating conditions are provided in the Supplementary material.

4.4. Addition of mixed butanols to the base gasoline fuels

The TPRF mixtures tested in the previous section are the surrogates of the Saudi Arabian gasoline fuels without the MTBE content (Refer Table 4 for the detailed composition). To further analyze the effect of mixed butanols addition to the Saudi Arabian oxygenate-free gasoline, the RON and MON values for the pure base gasoline (Gasolines A and B) and their blends with 10% and 20% mixed butanols were measured using the CFR engine facility at Saudi Aramco. In addition, the octane numbers of the pure TPRF-A and TPRF-B and their blends with 10% and 20% mixed butanol were measured and compared with the real gasoline data. All the results from the CFR engine facility are listed in Table 6. Octane ratings of the TPRFs and the base gasoline blended with mixed butanols.

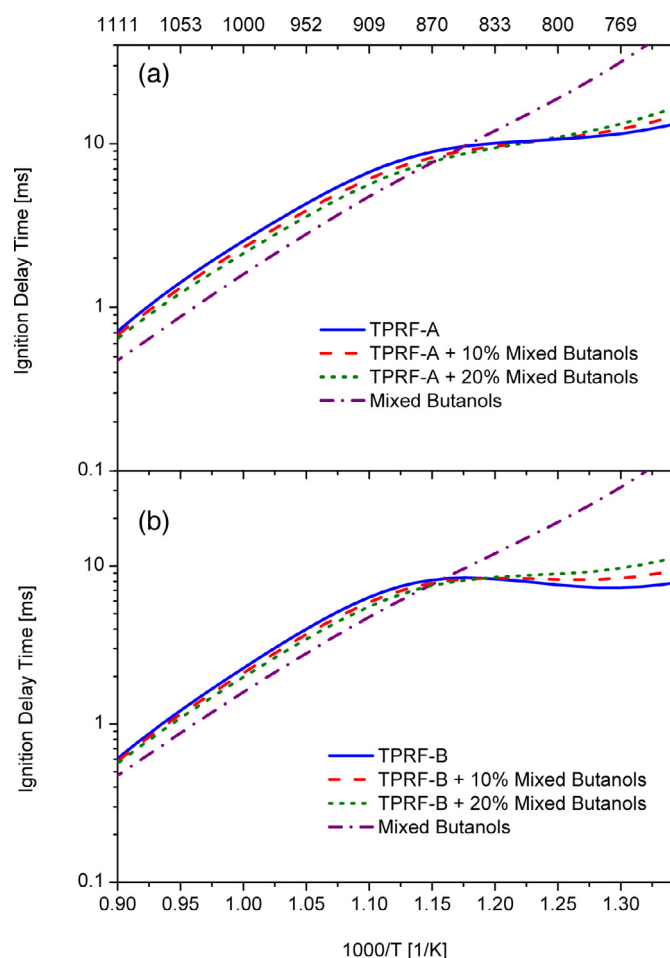


Fig. 7. Simulated ignition delay times of pure and mixed butanols + toluene/iso-octane/n-heptane mixtures at $P = 20$ bar, $\phi = 1$, (a) TPRF-A and (b) TPRF-B.

Table 6

Octane ratings of the TPRFs and the base gasoline blended with mixed butanols.

	Pure		+10% mixed butanols		+20% mixed butanols	
	RON	MON	RON	MON	RON	MON
Gasoline-A	90.0	81.7	91.5	83.5	93.5	84.4
TPRF-A	90.1	81.6	92.4	82.0	94.4	84.0
Gasoline-B	84.6	79.3	86.6	80.9	89.4	82.6
TPRF-B	84.6	79.1	89.3	80.0	91.0	83.5

As can be seen from Table 6, Octane ratings of the TPRFs and the base gasoline blended with mixed butanols, the TPRF surrogate model developed by Kalghatgi et al. [43] successfully reproduces the RON and MON of the base gasoline fuels. Secondly, the octane number response due to the addition of mixed butanols to the TPRF surrogates is larger than that of the base gasoline fuels. This difference in the octane number response between TPRF surrogates and the gasoline fuels is attributed to the compositional structure of the gasoline as compared to the TPRF surrogate. Foong et al. [46] showed that different TPRF mixtures with the same RON have different octane number responses when blended with ethanol. Since the octane number response is higher for the TPRF surrogates, it is expected that the difference in the ignition delay times of pure TPRF and TPRF + mixed butanols will be larger than that of pure gasoline and gasoline + mixed butanols due to the larger difference in the octane number. In addition, as demonstrated in Fig. 6, the addition of mixed butanols to TPRF has very marginal effect on the measured and simulated ignition delay times at the studied temperature range. Consequently, performing addition ignition delay time experiments for the base gasoline fuels

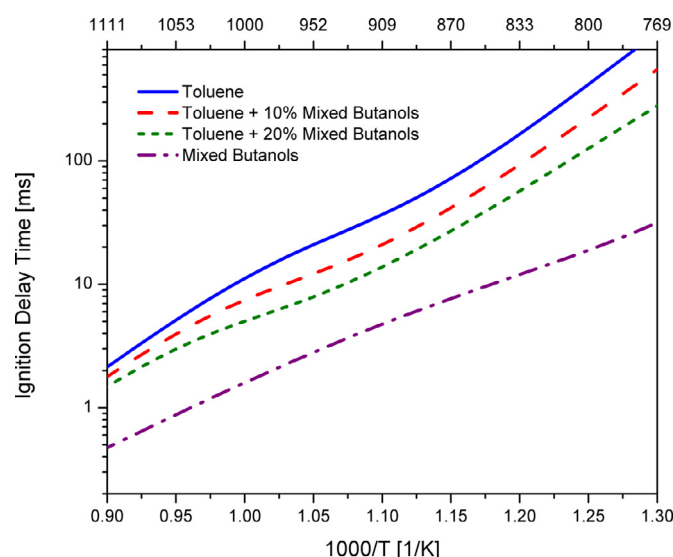


Fig. 8. Simulated ignition delay times of toluene and toluene/mixed butanols blends at $P = 20$ bar, $\phi = 1$.

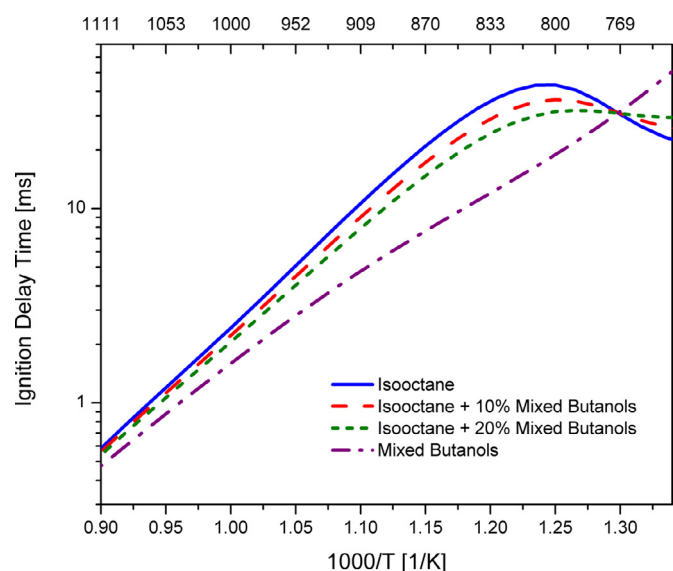


Fig. 9. Simulated ignition delay times of iso-octane and iso-octane/mixed butanols blends at $P = 20$ bar, $\phi = 1$.

and their mixtures with mixed butanols will not add any additional value to the current study.

4.5. Addition of mixed butanols to toluene and isooctane

As demonstrated in Section 4.3, mixed butanols, with RON of 106.7 and MON 94, act as octane boosters at low temperatures when blended with the TPRFs tested in this study. The addition of mixed butanols to toluene is investigated numerically by simulating the ignition delay times at 20 bar. Figure 8 shows the simulated ignition delay times for pure toluene (RON 120, MON 109), mixed butanols and toluene blended with 10% and 20% mixed butanols. The model indicates that the addition of mixed butanols increases the reactivity over the entire temperature range. Since toluene has a higher octane rating than mixed butanols, the addition of mixed butanols to toluene reduces the octane number of the toluene/mixed butanols blends.

Blending of iso-octane, which has RON and MON of 100 by definition, with mixed butanols is interesting because iso-octane's octane ratings lie between the RON and MON of mixed butanols. Figure 9

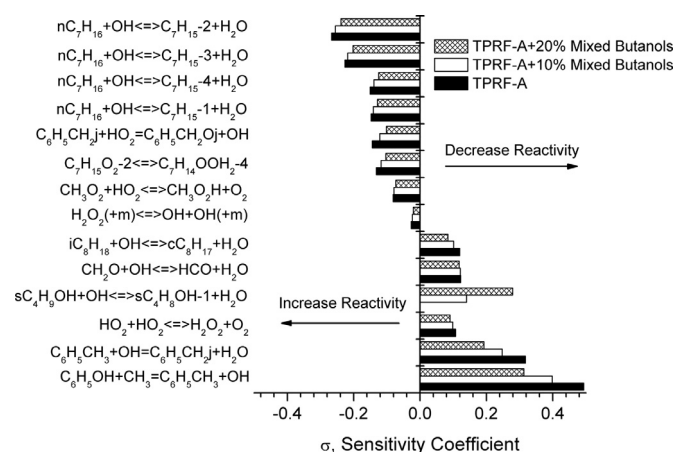


Fig. 10. Brute force sensitivity analysis of TPRF-A + Mixed butanols at 20 bar and 750 K.

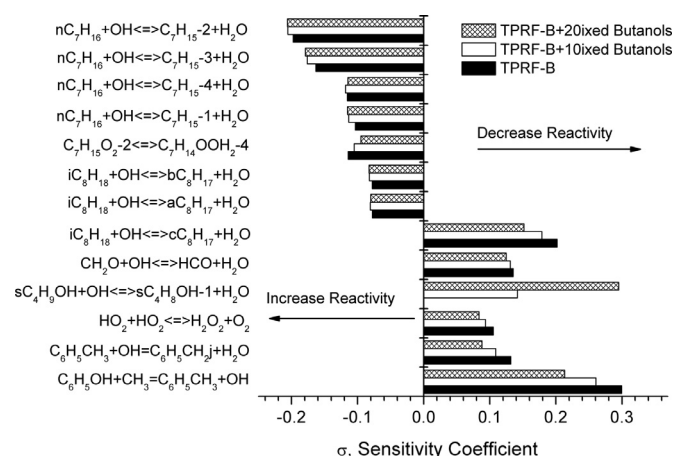


Fig. 12. Brute force sensitivity analysis of TPRF-B + Mixed butanols at 20 bar and 750 K.

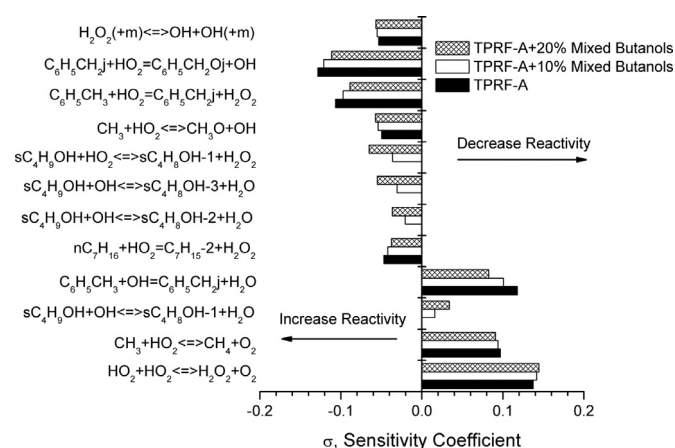


Fig. 11. Brute force sensitivity analysis of TPRF-A + Mixed butanols at 20 bar and 950 K.

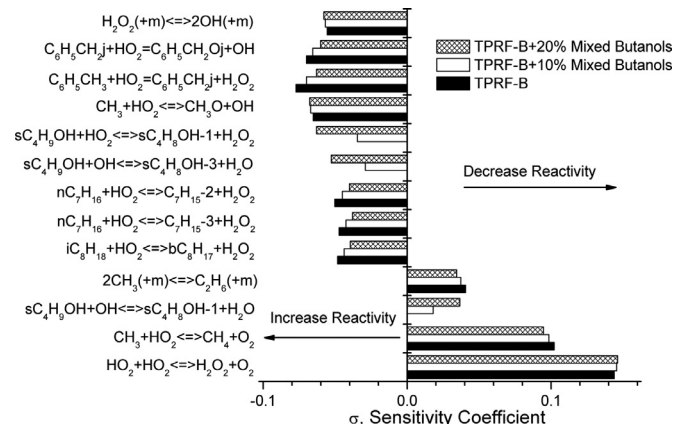


Fig. 13. Brute force sensitivity analysis of TPRF-B + Mixed butanols at 20 bar and 950 K.

shows the calculated ignition delay times of iso-octane, mixed butanols and iso-octane blended with 10% and 20% mixed butanols. Two distinct regions are observed. At $T < 770$ K, the mixture tends to resist auto-ignition with the addition of mixed butanols to iso-octane. In contrast, the reactivity of the mixture increases with the addition of mixed butanols to iso-octane at $T > 770$ K. This means that mixed butanols act as octane booster at temperatures lower than 770 K and as an octane retarder for temperatures higher than 770 K. By examining the RON and MON for both iso-octane and mixed butanols, one can assume that when the CFR engine is in RON mode, the representative temperature of the mixture should be below 770 K at 20 bar. The representative temperature should be above 770 K at 20 bar when the CFR engine is operating under MON conditions. The fact that the representative temperature for MON mode is higher than that for RON conditions is expected because the mixture temperature is heated to 149 °C in MON test compared to 52 °C under RON conditions.

4.6. Sensitivity analysis

Detailed analysis was carried out to identify sensitive reactions involved in the combustion of mixed butanols and gasoline surrogate mixtures. CHEMKIN-PRO was used to carry out brute force sensitivity analysis on ignition delay times by varying the A-factor of important reactions. The relevant reactions were identified by initially performing a temperature A-factor sensitivity analysis. The sensitivity

coefficients of selected reactions are then computed by the following formula,

$$\sigma = (\tau_+ - \tau) / \tau \quad (2)$$

where τ_+ is the computed ignition delay times corresponding to an increase by a factor two of the rate coefficient of the respective reaction and τ is the nominal ignition delay time. The analysis was performed for TPRF-A and B, TPRF-A and B + 10% mixed butanols and TPRF-A and B + 20% mixed butanols at 750 K and 950 K. The reason behind selecting these temperatures is that blending mixed butanols with TPRF-A and B decreases the reactivity at 750 K and increases the reactivity at 950 K (Refer to Figs. 7 and 8). Figures. 10–13 present the sensitivity analyses of TPRF-A at 750 K, TPRF-A at 950 K, TPRF-B at 750 K and TPRF-B at 950 K, respectively. As can be observed from Fig. 10, 2-butanol + OH alpha-site abstraction reaction plays a major role in reducing the reactivity of the fuel at 750 K, particularly at higher mixed butanols concentrations. As discussed by Sarathy et al. [25], abstraction from the alpha site of a 2-butanol always decreases the system reactivity at low temperatures because subsequent $R + O_2$ chain branching reactions are suppressed by the competing reaction of alpha-hydroxybutyl with O_2 to form 2-butanone + HO_2 . This clarifies the simulated results where the addition of mixed butanols increases the ignition delay time. At 750 K, n-heptane + OH reactions are promoting ignition and their effect decreases when adding mixed butanols. This behavior is expected because n-heptane concentration decreases with mixed butanols addition and hence

its contribution decreases. The addition of mixed butanols (by replacing some n-heptane) makes the primary oxidation pathway (n-heptane + OH) of lower importance in controlling the system reactivity. At 950 K, the 2-butanol + OH reaction trend is similar to what was observed at 750 K, however its magnitude is significantly smaller. This reaction is less important at 950 K because ignition is more controlled by high temperature radical beta-scissions reactions rather than R+O₂ low temperature reactions sequences. The toluene related reaction, $C_6H_5CH_3 + OH \rightleftharpoons C_6H_5CH_2j + H_2O$, exhibits larger sensitivity and are more controlling of ignition. The increase in reactivity with mixed butanols addition to TPRF-A at 950 K can be explained by the decreased sensitivity of the previously mentioned toluene reaction with the addition of mixed butanols, as can be seen from the lower part of Fig. 11.

Figure 12 shows the sensitivity analyses for TPRF-B + mixed butanols at 750 K. Similar behavior to what was described with TPRF-A sensitivity analyses at 750 K was observed here. n-heptane + OH reactions play a major role in promoting ignition and their contribution slightly increase when adding mixed butanols. 2-butanol + OH alpha-site abstraction reaction also plays a major role in inhibiting ignition at 750 K and its contribution significantly increases when adding mixed butanols. However, the ignition delay time sensitivity to 2-butanol + OH reaction is now larger than the $C_6H_5CH_3 + OH \rightleftharpoons C_6H_5CH_2j + H_2O$ and $C_6H_5OH + CH_3 \rightleftharpoons C_6H_5CH_3 + OH$ reactions. This was expected because of the lower toluene concentration in TPRF-B (28.8 vol%) compared to TPRF-A (49.4 vol%). The TPRF-B sensitivity analyses at 950 K, as shown in Fig. 13, demonstrate no dependence on $C_6H_5CH_3 + OH \rightleftharpoons C_6H_5CH_2j + H_2O$ to decrease the reactivity, in which TPRF-A sensitivity analyses at 950 K showed slight dependence. This behavior is expected because of the lower toluene content of TPRF-B when compared to TPRF-A. In addition, even though 2-butanol + OH reaction decrease the reactivity of the mixture, 2-butanol + HO₂ reaction plays greater role in promoting reactivity and its effect increases with mixed butanols addition. Therefore, the addition of mixed butanols to TPRF-B promotes ignition at 950 K.

5. Conclusions

A chemical kinetic mechanism was developed for mixed butanols (2 butanol / tert-butanol) and toluene primary reference fuel (TPRF) surrogates by merging sub-mechanisms available in the literature. To complement the model, ignition delay time measurements were performed for mixed butanols and for two TPRF mixtures blended with mixed butanols. The proposed model is in good agreement with the obtained experimental data and previous measurements reported in literature. The addition of mixed butanols to TPRF surrogates can either increase or decrease the reactivity depending on the temperature range. The crossover temperature is around 850 K; at lower temperatures, the addition of mixed butanols increases the ignition delay times and the opposite trend is observed at higher temperatures. Analysis performed using the developed chemical kinetic mechanism revealed that mixed butanols addition to pure toluene always speeds up reactions, whereas, addition of mixed butanols to pure iso-octane results in similar behavior to that of TPRF where a crossover is observed at around 770 K. This study showed that mixed butanols can potentially be used as octane boosters in spark ignition engines.

Acknowledgments

Work reported in this paper was funded by Saudi Aramco and King Abdullah University of Science and Technology (KAUST) under FUELCOM program. We also acknowledge competitive research funding from the KAUST Clean Combustion Research Center under the Future Fuels research program.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2015.07.035.

References

- [1] IEA (2011), Deploying Renewables 2011, Best and Future Policy Practices, OECD Publishing Paris, <http://dx.doi.org/10.1787/9789264124912-en>.
- [2] D. Petitjean, L. Bernardini, C. Middlemass, S. Shahed, SAE Technical Paper 2004-01-0988 (2004).
- [3] J. Milpied, N. Jeuland, G. Plassat, S. Guichaou, N. Dioc, A. Marchal, P. Schmelzle, SAE Int. J. Fuels Lubr 2 (1) (2009) 118–126.
- [4] S. Shahed, K.-H. Bauer, SAE Int. J. Engines 2 (1) (2009) 1347–1358.
- [5] G. Kalghatgi, Fuel/engineInteractions, SAE International, Warrendale, PA, 2013, pp. 109–128.
- [6] G.T. Kalghatgi, Int. J. Engine Res. (2014), doi:10.1177/1468087414526189.
- [7] Standard Test Method for Research Octane Number of Spark-Ignition Engine Fuel, 2013; vol. ASTM D2699 - 13b.
- [8] Standard Test Method for Motor Octane Number of Spark-Ignition Engine Fuel, 2014; vol. ASTM D 2700-14.
- [9] S.M. Sarathy, P. Oßwald, N. Hansen, K. Kohse-Höinghaus, Progress Energy Combust. Sci. 44 (2014) 40–102.
- [10] P.H. Pfromm, V. Amanor-Boadu, R. Nelson, P. Vadlani, R. Madl, Biomass Bioenergy 34 (4) (2010) 515–524.
- [11] K. Bullis BP's Beton Butanol <http://www.technologyreview.com/news/407591/459bbs-bet-on-butanol/> (accessed on August 5, 2015).
- [12] NACS Butanol vs. Ethanol <http://www.nacsonline.com/news/daily/pages/461ND041513.aspx#.VNGywiUeAU> (accessed on August 5, 2015).
- [13] P.S. Nigam, A. Singh, Progress Energy Combust. Sci. 37 (1) (2011) 52–68.
- [14] I. Stranic, D.P. Chase, J.T. Harmon, S. Yang, D.F. Davidson, R.K. Hanson, Combust. Flame 159 (2) (2012) 516–527.
- [15] I. Stranic, S.H. Pyun, D.F. Davidson, R.K. Hanson, Combust. Flame 159 (11) (2012) 3242–3250.
- [16] K.E. Noorani, B. Akih-Kumgeh, J.M. Bergthorson, Energy Fuels 24 (11) (2010) 5834–5843.
- [17] T. Choudhury, M. Lna, C.-Y. Lin, W. Sanders, Combust. Sci. Technol. 71 (4–6) (1990) 219–232.
- [18] J.T. Moss, A.M. Berkowitz, M.A. Oehlschlaeger, J. Biet, V. Warth, P.-A. Glaude, F. Battin-Leclerc, J. Phys. Chem. A 112 (43) (2008) 10843–10855.
- [19] I. Stranic, S.H. Pyun, D.F. Davidson, R.K. Hanson, Combust. Flame 160 (6) (2013) 1012–1019.
- [20] S.S. Vasu, S.M. Sarathy, Energy Fuels 27 (11) (2013) 7072–7080.
- [21] Y. Zhu, D.F. Davidson, R.K. Hanson, Combust. Flame 161 (3) (2014) 634–643.
- [22] G. Black, H. Curran, S. Pichon, J. Simmie, V. Zhukov, Combust. Flame 157 (2) (2010) 363–373.
- [23] K. Heufer, R. Fernandes, H. Olivier, J. Beeckmann, O. Röhl, N. Peters, Proc. Combust. Instit. 33 (1) (2011) 359–366.
- [24] S. Vranckx, K. Heufer, C. Lee, H. Olivier, L. Schill, W. Kopp, K. Leonhard, C. Taatjes, R. Fernandes, Combust. Flame 158 (8) (2011) 1444–1455.
- [25] S.M. Sarathy, S. Vranckx, K. Yasunaga, M. Mehl, P. Oßwald, W.K. Metcalfe, C.K. Westbrook, W.J. Pitz, K. Kohse-Höinghaus, R.X. Fernandes, Combust. Flame 159 (6) (2012) 2028–2055.
- [26] M.R. Harper, K.M. Van Geem, S.P. Pyl, G.B. Marin, W.H. Green, Combust. Flame 158 (1) (2011) 16–41.
- [27] S. Sarathy, M. Thomson, C. Togbé, P. Dagaut, F. Halter, C. Mounaim-Rousselle, Combust. Flame 156 (4) (2009) 852–864.
- [28] P.S. Veloo, Y.L. Wang, F.N. Egolfopoulos, C.K. Westbrook, Combust. Flame 157 (10) (2010) 1989–2004.
- [29] R. Grana, A. Frassoldati, T. Faravelli, U. Niemann, E. Ranzi, R. Seiser, R. Cattolica, K. Seshadri, Combust. Flame 157 (11) (2010) 2137–2154.
- [30] W. Xu, S.R. Vogel, Combustible Mixed Butanol Fuels, U.S. Patent Application 12/824,432.
- [31] W. Xu, F.M. Al-Shahrani, A. Bourane, S.R. Vogel, Process for the hydration of mixed butenes to produce mixed alcohols, U.S. Patent Application 13/977,860.
- [32] T. Niass, A. Amer, W. Xu, S. Vogel, K. Krebber-Hortmann, P. Adomeit, A. Brassat, SAE Int. J. Fuels Lubr. 5 (1) (2011) 265–273.
- [33] S.M. Burke, U. Burke, R. Mc Donagh, O. Mathieu, I. Osorio, C. Keesee, A. Morones, E.L. Petersen, W. Wang, T.A. DeVerter, Combust. Flame 162 (2) (2015) 296–314.
- [34] A.R. Amadio, M.W. Crofton, E.L. Petersen, Shock Waves 16 (2) (2006) 157–165.
- [35] D. Davidson, R. Hanson, Int. J. Chem. Kinet. 36 (9) (2004) 510–523.
- [36] W.K. Metcalfe, S.M. Burke, S.S. Ahmed, H.J. Curran, Int. J. Chem. Kinet. 45 (10) (2013) 638–675.
- [37] M. Mehl, W.J. Pitz, C.K. Westbrook, H.J. Curran, Proc. Combust. Instit. 33 (1) (2011) 193–200.
- [38] M.R. McGillen, M. Baasandorj, J.B. Burkholder, J. Phys. Chem. A 117 (22) (2013) 4636–4656.
- [39] CHEMKIN-PRO 15112, Reaction Design, San Diego, 2012.
- [40] M. Chaos, F.L. Dryer, Int. J. Chem. Kinet. 42 (3) (2010) 143–150.
- [41] B.W. Weber, C.-J. Sung, Energy Fuels 27 (3) (2013) 1688–1698.
- [42] B.W. Weber, K. Kumar, Y. Zhang, C.-J. Sung, Combust. Flame 158 (5) (2011) 809–819.
- [43] G. Kalghatgi, H. Babiker, J. Badra, SAE Technical Paper 2015-01-0757 (2015).
- [44] M. Mehl, T. Faravelli, F. Giavazzi, E. Ranzi, P. Scorletti, A. Tardani, D. Terna, Energy Fuels 20 (6) (2006) 2391–2398.

- [45] S.M. Sarathy, T. Javed, F. Karsenty, A. Heufer, W. Wang, S. Park, A. Elwardany, A. Farooq, C.K. Westbrook, W.J. Pitz, *Combust. Flame* 161 (6) (2014) 1444–1459.
- [46] T.M. Foong, K.J. Morganti, M.J. Brear, G. da Silva, Y. Yang, F.L. Dryer, *Fuel* 115 (2014) 727–739.
- [47] E. Christensen, J. Yanowitz, M. Ratcliff, R.L. McCormick, *Energy Fuels* 25 (10) (2011) 4723–4733.
- [48] A. Sharif, *Tikrit J. Eng. Sci.* 17 (2) (2010) 22–35.
- [49] N. Morgan, A. Smallbone, A. Bhave, M. Kraft, R. Cracknell, G. Kalghatgi, *Combust. Flame* 157 (6) (2010) 1122–1131.